PROBLEM 08

Kinematics, Two Trains on Tracks

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Relative Motion

Two trains, A and B, are moving along parallel tracks. The separation between the tracks is a. The absolute velocity of each train is v_A and v_B, respectively. What is the relative velocity of each train with respect to the other?

Solve the same problem considering that the train B is now moving along a circular track of radius R. The minimum distance between the circular track and the straight track of A is a.

Worked Solution

Reference Frames, Bases and Coordinates

- **Physical System:** Two trains A and B move on separate tracks with distance of closest separation a.
- Case 1: Two parallel straight tracks with constant speeds v_A and v_B.
- Case 2: Train A on a straight track, Train B on a circular track of radius R.
- **Parameters & Units:** Separation a $[\mathrm{m}]$, radius R $[\mathrm{m}]$, speeds v_A, v_B $[\mathrm{m/s}]$.
- **Hypotheses:** the tracks are in a plane; at $t = 0$ train A is at the origin and train B is at the point of its track closest to the track of A; the speeds are constant.
- **Inertial Cartesian Frame:** $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}\}$, with $\mathbf{i}_0$ along the track of A, $\mathbf{j}_0$ towards the track of B and $\mathbf{k} = \mathbf{i}_0 \times \mathbf{j}_0$ normal to the plane of the tracks.
- **Frame of train B (Case 2, rotating observer):** $S_B = \{B; \mathbf{i}_B, \mathbf{j}_B, \mathbf{k}\}$, attached to the train, with $\mathbf{i}_B$ along its direction of motion (the tangent to the track) and $\mathbf{j}_B$ pointing towards the centre C of the track. It turns with the train by the angle $\omega_B t$ about $\mathbf{k}$, $\omega_B = v_B/R$.

- **Coordinates and degrees of freedom.** The coordinates are the arc positions along the tracks (one per train). Both motions are prescribed, $x_A = v_A t$ and $\theta_B = v_B t / R$, so no degree of freedom is unknown (0 DOF): the problem is purely kinematic. θ_B is the angle of CB measured at the centre C from the downward vertical $-\mathbf{j}_0$, positive counter-clockwise (towards $+\mathbf{i}_0$).
- **Observers.** In Case 1 the relative velocity is the same for any observer that translates with a train. In Case 2 an observer attached to B turns with it at $\omega_B = v_B / R$, so the velocity of A seen by this observer needs the rotating frame S_B.

Change of Basis and Position Vector

Position of each train

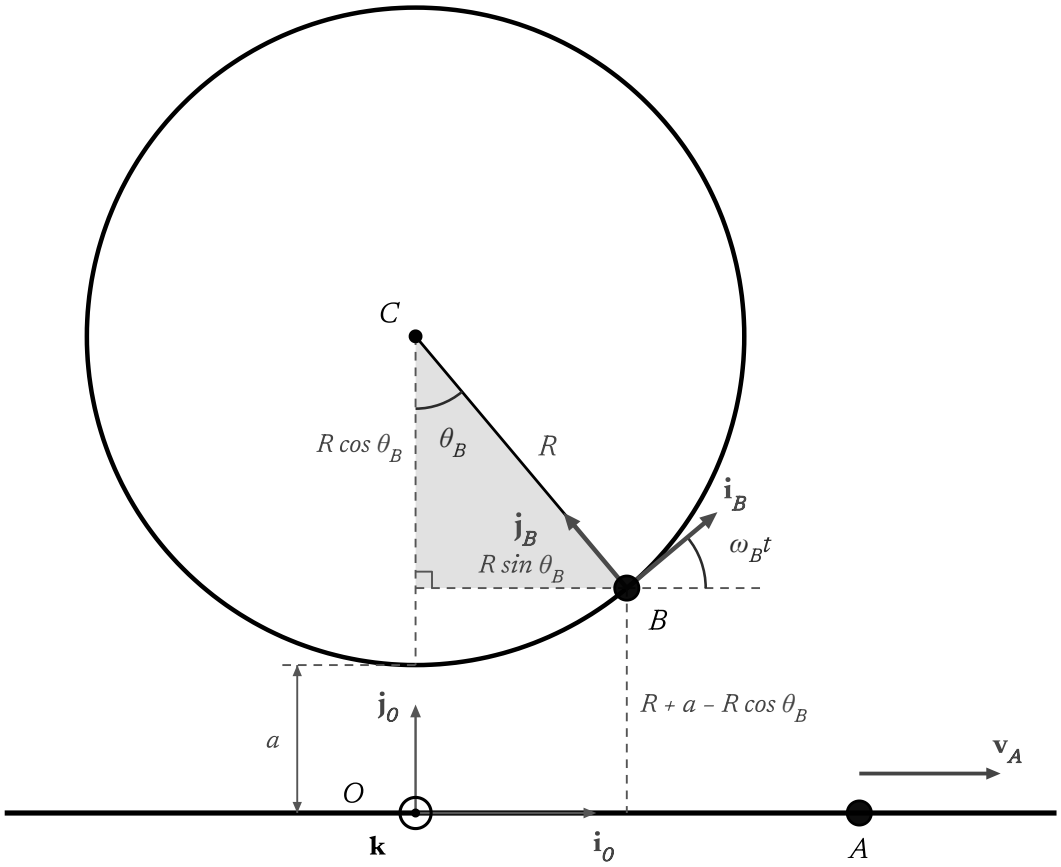


Figure 1. Plane of the tracks ($\mathbf{k}$ out of the paper). $\mathbf{r}_B = OC + CB$: in the shaded right triangle CFB the hypotenuse is $|CB| = R$, the leg opposite θ_B is $R\sin\theta_B$ (along $\mathbf{i}_0$) and the adjacent leg is $R\cos\theta_B$ (along $-\mathbf{j}_0$). The tangent $\mathbf{i}_B$ makes the same angle $\theta_B = \omega_B t$ with the horizontal; $\mathbf{j}_B$ points to C.

The plane of the tracks is drawn from above: $\mathbf{k}$ points out of the paper, towards the reader ($\odot$), $\mathbf{i}_0$ to the right and $\mathbf{j}_0$ up (the right-hand rule $\mathbf{i}_0 \times \mathbf{j}_0 = \mathbf{k}$: fingers from $\mathbf{i}_0$ to $\mathbf{j}_0$ counter-clockwise, thumb towards the reader).

Case 1. Train A runs on the line $y = 0$ and train B on the parallel line $y = a$:

$$\mathbf{r}_A(t) = v_A t\,\mathbf{i}_0, \qquad \mathbf{r}_B(t) = (x_{B0} + v_B t)\,\mathbf{i}_0 + a\,\mathbf{j}_0$$

Case 2. Train A is still on the x-axis, $\mathbf{r}_A = v_A t\,\mathbf{i}_0$. The circular track has its centre C at $(0, R + a)$, so that its lowest point is at height $(R + a) - R = a$ above the track of A, the given minimum distance. To reach B we pass through the centre, because the distance from C is the one that stays constant:

$$\mathbf{r}_B = \overrightarrow{OC} + \overrightarrow{CB}$$

- $\overrightarrow{OC} = (R + a)\,\mathbf{j}_0$.
- $\overrightarrow{CB}$ has length R and makes the angle $\theta_B = \omega_B t$ with the downward vertical. Let F be the point of the vertical through C at the height of B. In the shaded right triangle CFB (right angle at F) the hypotenuse is $|CB| = R$; the leg opposite to θ_B is $FB = R\sin\omega_B t$, horizontal, along $+\mathbf{i}_0$; the leg adjacent to θ_B is $CF = R\cos\omega_B t$, vertical and pointing down, along $-\mathbf{j}_0$. So $\overrightarrow{CB} = R\sin\omega_B t\,\mathbf{i}_0 - R\cos\omega_B t\,\mathbf{j}_0$.

Adding:

$$\mathbf{r}_B(t) = R\sin(\omega_B t)\,\mathbf{i}_0 + [R + a - R\cos(\omega_B t)]\,\mathbf{j}_0$$

Check: $t = 0$ gives $a\,\mathbf{j}_0$, the lowest point of the circle, closest to the track of A. The angle θ_B is measured from $-\mathbf{j}_0$ and not from $\mathbf{i}_0$: the usual polar angle about C is $\varphi = \theta_B - \pi/2$, because $\cos\left(\theta_B - \frac{\pi}{2}\right) = \cos\theta_B \cos\frac{\pi}{2} + \sin\theta_B \sin\frac{\pi}{2} = \sin\theta_B$ and $\sin\left(\theta_B - \frac{\pi}{2}\right) = \sin\theta_B \cos\frac{\pi}{2} - \cos\theta_B \sin\frac{\pi}{2} = -\cos\theta_B$, which turns $R(\cos\varphi\,\mathbf{i}_0 + \sin\varphi\,\mathbf{j}_0)$ into the $\overrightarrow{CB}$ above. It is chosen so that $\theta_B = 0$ is the starting point.

The basis of train B

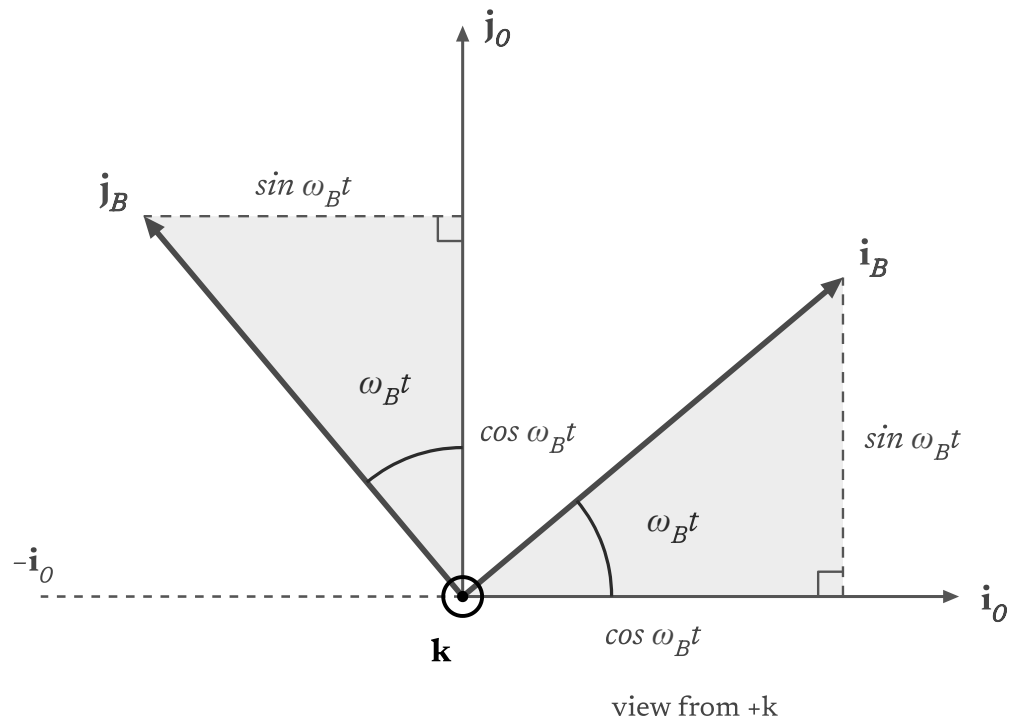


Figure 2. View from $+\mathbf{k}$ ($\odot$). $\mathbf{i}_B$ is $\mathbf{i}_0$ turned counter-clockwise by $\omega_B t$: legs $\cos\omega_B t$ along $\mathbf{i}_0$ and $\sin\omega_B t$ along $\mathbf{j}_0$. $\mathbf{j}_B$ turns the same angle from $\mathbf{j}_0$, so its opposite leg $\sin\omega_B t$ points along $-\mathbf{i}_0$ (the minus sign of $[{}_0R_B]$).

$\mathbf{i}_B$ is the unit tangent to the track in the sense of motion. Differentiating $\overrightarrow{CB}$ with respect to θ_B ($\frac{d}{d\theta}\sin\theta = \cos\theta$, $\frac{d}{d\theta}(-\cos\theta) = \sin\theta$) and dividing by its length $R\sqrt{\cos^2\theta_B + \sin^2\theta_B} = R$:

$$\mathbf{i}_B = \cos\omega_B t\,\mathbf{i}_0 + \sin\omega_B t\,\mathbf{j}_0$$

so $\mathbf{i}_B$ is $\mathbf{i}_0$ turned counter-clockwise by $\omega_B t$ (seen from $+\mathbf{k}$, $\odot$): leg $\cos\omega_B t$ along $\mathbf{i}_0$ (adjacent) and leg $\sin\omega_B t$ along $\mathbf{j}_0$ (opposite). $\mathbf{j}_B = \mathbf{k}\times\mathbf{i}_B$ is $\mathbf{i}_B$ turned a further 90°; it has turned the same angle $\omega_B t$ from $\mathbf{j}_0$, and turning counter-clockwise from "up" moves its tip towards $-\mathbf{i}_0$, so its opposite leg is $-\sin\omega_B t\,\mathbf{i}_0$:

$$\mathbf{j}_B = -\sin\omega_B t\,\mathbf{i}_0 + \cos\omega_B t\,\mathbf{j}_0 = -\frac{\overrightarrow{CB}}{R}$$

(comparing with $\overrightarrow{CB} = R\sin\omega_B t\,\mathbf{i}_0 - R\cos\omega_B t\,\mathbf{j}_0$), so $\mathbf{j}_B$ points from B towards the centre C, as drawn. With the columns of $\mathbf{i}_B, \mathbf{j}_B, \mathbf{k}$ in $\mathcal{B}_0$:

$$[{}_0R_B] = \begin{pmatrix} \cos\omega_B t & -\sin\omega_B t & 0 \\ \sin\omega_B t & \cos\omega_B t & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_B] = \cos\omega_B t\cos\omega_B t - (-\sin\omega_B t)\sin\omega_B t = \cos^2\omega_B t + \sin^2\omega_B t = 1$$

and $[{}_0R_B][{}_0R_B]^T$ has diagonal entries $\cos^2\omega_B t + \sin^2\omega_B t = 1$ (and 1) and off-diagonal entries $\cos\omega_B t\sin\omega_B t - \sin\omega_B t\cos\omega_B t = 0$, so it is I. The angular velocity of S_B is $\boldsymbol{\omega}_{B0} = \omega_B\,\mathbf{k}$ (Poisson: $\frac{d\mathbf{i}_B}{dt} = -\omega_B\sin\omega_B t\,\mathbf{i}_0 + \omega_B\cos\omega_B t\,\mathbf{j}_0 = \omega_B\,\mathbf{j}_B = \omega_B\,\mathbf{k}\times\mathbf{i}_B$).

Kinematics

Case 1: Parallel Tracks

Differentiating the positions (the speeds are constant):

$$\mathbf{v}_A = v_A\,\mathbf{i}_0, \qquad \mathbf{v}_B = v_B\,\mathbf{i}_0$$

- **Relative Velocity:**

$$\mathbf{v}_{B/A} = \mathbf{v}_B - \mathbf{v}_A = (v_B - v_A)\mathbf{i}_0$$

$$\mathbf{v}_{A/B} = -\mathbf{v}_{B/A} = (v_A - v_B)\mathbf{i}_0$$

Case 2: Train B on a Circular Track of Radius R

- Velocity of train B, with the chain rule $\frac{d}{dt}\sin(\omega_B t) = \omega_B\cos(\omega_B t)$ and $\frac{d}{dt}\cos(\omega_B t) = -\omega_B\sin(\omega_B t)$:

$$\mathbf{v}_B = \frac{d\mathbf{r}_B}{dt} = R\omega_B\cos(\omega_B t)\mathbf{i}_0 + R\omega_B\sin(\omega_B t)\mathbf{j}_0 = v_B\cos\left(\frac{v_B t}{R}\right)\mathbf{i}_0 + v_B\sin\left(\frac{v_B t}{R}\right)\mathbf{j}_0 = v_B\,\mathbf{i}_B$$

- **Relative Velocity $\mathbf{v}_{B/A}$:**

$$\mathbf{v}_{B/A} = \mathbf{v}_B - \mathbf{v}_A = \left[v_B\cos\left(\frac{v_B t}{R}\right) - v_A\right]\mathbf{i}_0 + v_B\sin\left(\frac{v_B t}{R}\right)\mathbf{j}_0$$

- **Magnitude of Relative Velocity** (expanding the square and using $\cos^2 + \sin^2 = 1$):

$$v_{\text{rel}} = \sqrt{\left[v_B\cos\left(\frac{v_B t}{R}\right) - v_A\right]^2 + v_B^2\sin^2\left(\frac{v_B t}{R}\right)} = \sqrt{v_B^2 + v_A^2 - 2v_A v_B\cos\left(\frac{v_B t}{R}\right)}$$

Case 2, velocity of A with respect to an observer attached to B

The frame S_B of the change-of-basis section turns with $\boldsymbol{\omega}_{B0} = \omega_B\mathbf{k}$. Notes Eq. 3.26 gives $\mathbf{v}_{A/B} = \mathbf{v}_A - \mathbf{v}_B - \boldsymbol{\omega}_{B0}\times(\mathbf{r}_A - \mathbf{r}_B)$. With $\mathbf{r}_A - \mathbf{r}_B = (v_A t - R\sin\omega_B t)\,\mathbf{i}_0 + (R\cos\omega_B t - R - a)\,\mathbf{j}_0$, $\mathbf{k}\times\mathbf{i}_0 = \mathbf{j}_0$, $\mathbf{k}\times\mathbf{j}_0 = -\mathbf{i}_0$ and $\omega_B R = v_B$, the three terms are

$$\mathbf{v}_A - \mathbf{v}_B = (v_A - v_B\cos\omega_B t)\,\mathbf{i}_0 - v_B\sin\omega_B t\,\mathbf{j}_0, \qquad -\boldsymbol{\omega}_{B0}\times(\mathbf{r}_A - \mathbf{r}_B) = -\omega_B\,(R + a - R\cos\omega_B t)\,\mathbf{i}_0 - \omega_B\,(v_A t - R\sin\omega_B t)\,\mathbf{j}_0$$

Adding them, $\omega_B R\cos\omega_B t = v_B\cos\omega_B t$ cancels the $-v_B\cos\omega_B t$ term and $\omega_B R\sin\omega_B t = v_B\sin\omega_B t$ cancels the $-v_B\sin\omega_B t$ term:

$$\mathbf{v}_{A/B} = \left[v_A - v_B\,\frac{R+a}{R}\right]\mathbf{i}_0 - \frac{v_A v_B t}{R}\,\mathbf{j}_0 \quad (\text{components in } \mathcal{B}_0)$$

Answers and Checks

Case 1

$\mathbf{v}_{B/A} = (v_B - v_A)\,\mathbf{i}_0$ and $\mathbf{v}_{A/B} = (v_A - v_B)\,\mathbf{i}_0$.

Case 2

$\mathbf{v}_{B/A} = [v_B\cos(v_B t/R) - v_A]\,\mathbf{i}_0 + v_B\sin(v_B t/R)\,\mathbf{j}_0$, with modulus $v_{\text{rel}} = \sqrt{v_A^2 + v_B^2 - 2v_A v_B\cos(v_B t/R)}$, for an observer that translates with A ($\mathbf{v}_{A/B} = -\mathbf{v}_{B/A}$ for an observer that translates with B). No integration is needed beyond the differentiation above.

For an observer attached to B (turning with it), $\mathbf{v}_{A/B} = [v_A - v_B(R+a)/R]\,\mathbf{i}_0 - (v_A v_B t/R)\,\mathbf{j}_0$. It differs from $-\mathbf{v}_{B/A}$ because the observer attached to B rotates: at $t = 0$, $v_{A/B} = v_A - v_B(1 + a/R)$, whereas the translating-observer value is $v_A - v_B$; for $R \to \infty$ both coincide.

Checks

- **Dimensional check:** $[\mathbf{v}_{B/A}] = \mathrm{m/s}$; in Case 2 the argument of the trigonometric functions is $\dfrac{v_B t}{R} = \dfrac{(\mathrm{m/s})\,\mathrm{s}}{\mathrm{m}}$, which is dimensionless (rad), and $[v_B^2 + v_A^2 - 2v_A v_B\cos(\cdot)] = \mathrm{m^2/s^2}$, so v_{rel} is in m/s.
- **Role of the separation a:** it does not appear in any relative velocity for translating observers, because $\mathbf{v}_{B/A}$ is the time derivative of $\mathbf{r}_B - \mathbf{r}_A$ and a only enters the constant offset of that vector (it does appear for the rotating observer, through $\boldsymbol{\omega}_{B0} \times (\mathbf{r}_A - \mathbf{r}_B)$).
- **Limiting cases of Case 2 (sympy):**
- $t = 0$: $v_{\text{rel}} = \sqrt{v_A^2 + v_B^2 - 2v_A v_B} = |v_B - v_A|$, because both velocities are parallel to $\mathbf{i}_0$ at that instant.
- $R \to \infty$ (the circular track straightens): $\cos(v_B t/R) \to 1$ and $v_{\text{rel}} \to |v_B - v_A|$, which is the Case 1 result.
- $v_A = v_B = v$: $v_{\text{rel}} = v\sqrt{2 - 2\cos(vt/R)} = 2v\left|\sin\dfrac{vt}{2R}\right|$, which starts at 0 and never exceeds $2v$.
- $v_A = 0$: $v_{\text{rel}} = v_B$ for all t (relative to a stationary train, only the speed of B matters).

- **Symbolic check:** $\|\mathbf{v}_B - \mathbf{v}_A\|^2$ computed from the components equals $v_B^2 + v_A^2 - 2v_A v_B \cos(v_B t/R)$ with zero residual. The translating-observer results use only $\mathcal{B}_0$; the rotating observer of Case 2 uses $[{}_0R_B]$. Dimensions of that result: $[v_A] = [v_B a/R] = [v_A v_B t/R] = \mathrm{m/s}$.
- **Check against the official key:** `Problems.pdf` has no entry for Problem 8 in its solution keys, so the results are verified by independent derivation (the symbolic check above).

Source: Problems.pdf